

ELECENG 3TP3: SIGNALS AND SYSTEMS

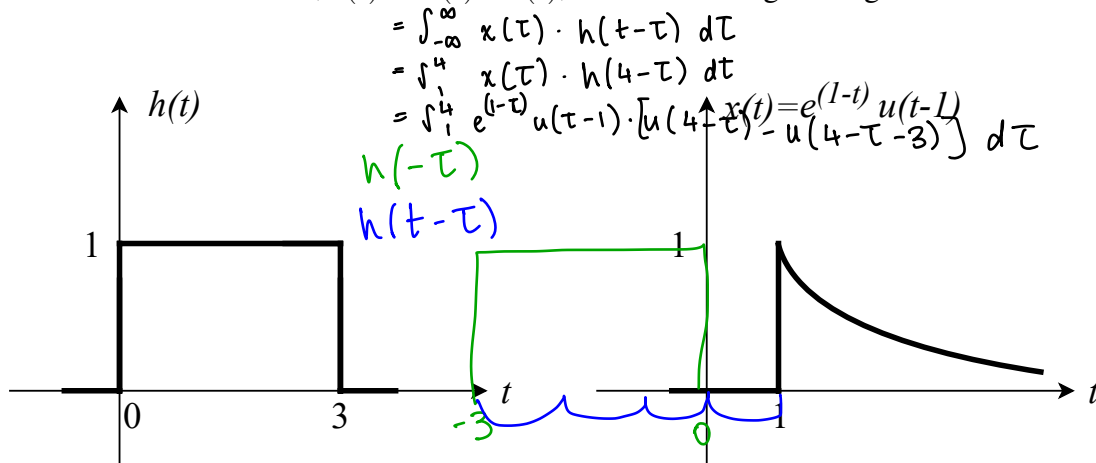
MIDTERM TEST 2023 (SOLUTIONS)

Instructions: 50 minute closed book test. One double-sided 8.5" × 11" formula sheet permitted. Answer all questions on the sheets provided. Please put your name on all sheets.

Name: _____ ID: _____

Section: _____

- (30) 1. Consider the convolution, $c(t) = x(t) * h(t)$, of the following two signals.



- (6) (a) Write an expression of $h(t)$ in terms of unit-step function.

$$h(t) = u(t) - u(t-3)$$

- (8) (b) For what range(s) of values of t is $c(t)$ equal to zero?

$$c(t) = 0 \text{ when } [-\infty < t \leq +1]$$

- (8) (c) Write out the integral(s) (including the integration limits) that can be evaluated in order to find $c(t)$ for $1 \leq t < 4$. Do not evaluate the expression. Draw a box around your final answer.

$$\int_1^4 e^{1-\tau} u(\tau-1) \cdot [u(4-\tau) - u(4-\tau-3)] d\tau$$

- (8) (d) Write out the integral(s) (including the integration limits) that can be evaluated in order to find $c(t)$ for $t \geq 4$. Do not evaluate the expression. Draw a box around your final answer.

$$\int_4^t e^{1-\tau} d\tau$$

$$\begin{aligned}
 v(t) &= \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\tau \\
 &= \int_{-\infty}^{\infty} e^{1-\tau} u(\tau-1) \cdot h(t-\tau) d\tau \\
 &= \int_1^t e^{1-\tau} \cdot h(t-\tau) d\tau \\
 &= \int_{t-3}^t e^{1-\tau} d\tau
 \end{aligned}$$

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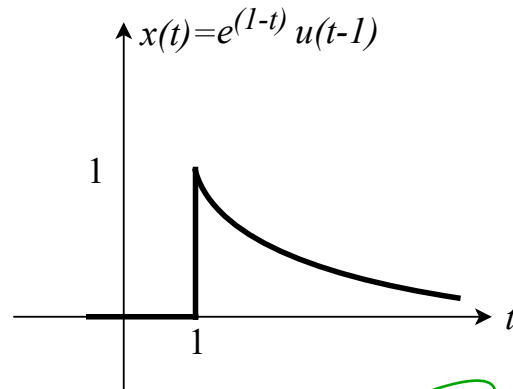
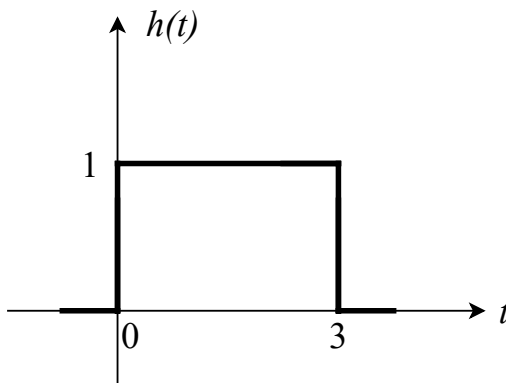
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- (30) 1. Consider the convolution, $c(t) = x(t) * h(t)$, of the following two signals.



$$\begin{aligned}
 c(t) &= x(t) * h(t) \\
 &= \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\tau \\
 &= \int_{-\infty}^{\infty} e^{(1-\tau)} u(\tau-1) \cdot h(t-\tau) d\tau \\
 &= \int_1^t e^{1-\tau} \cdot h(t-\tau) d\tau \\
 &= \int_1^t e^{1-\tau} \cdot 1 d\tau \longrightarrow \int_1^t e^{1-\tau} d\tau
 \end{aligned}$$

- (8) (c) Write out the integral(s) (including the integration limits) that can be evaluated in order to find $c(t)$ for $1 \leq t < 4$. Do not evaluate the expression. Draw a box around your final answer.

$\int_1^t e^{(1-\tau)} d\tau \text{ OR } \int_0^{t-1} e^{-(t-\tau-1)} d\tau$

- (10) 2. Determine whether the following discrete-time system is causal or noncausal, have memory or is memoryless, is linear or nonlinear, is time invariant or time varying. Justify your answers. In the following system, $x[n]$ is an arbitrary input and $y[n]$ is the response to $x[n]$.

$$y[n] = \frac{1}{2}x[n] + 3x[n-1] + \underbrace{y[n-1]}_{\text{causal}} + \underbrace{y[n+1]}_{\text{non causal}}$$

i) causal or non causal?

causal. the system only depends on the past or current value of the input.
 $\hookrightarrow$ ie. $x[a]$, where $a \leq n$.

ii) memory or memoryless?

memory. depends on previous values.

$$\hookrightarrow \text{eg. } y[3] = \frac{1}{2}x[3] + 3x[3-1] \\ = \frac{1}{2}x[3] + 3x[2] \longrightarrow y[3] \text{ depends on value of past: } x[2]$$

iii) (non) linear?

linear. it has both additivity & homogeneity.

$$\hookrightarrow \text{eg. } y[n] = \frac{1}{2}x[n] + 3x[n-1]$$

$$\text{additivity: } y[n+a] = \frac{1}{2}x[n+a] + 3x[n-1+a] \quad \checkmark$$

$$y_1[n] + y_2[n] = \frac{1}{2}x_1[n] + 3x_1[n-1] + \frac{1}{2}x_2[n] + 3x_2[n-1] \\ = \frac{1}{2}(x_1+x_2)[n] + 3(x_1+x_2)[n-1] \quad \checkmark$$

$$\text{homogeneity: } ay[n] = a\left(\frac{1}{2}x[n] + 3x[n-1]\right) \\ = \frac{a}{2}x[n] + 3ax[n-1] \quad \checkmark$$

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$$y[n] = \frac{1}{2}x[n] + 3x[n-1]$$

iv) time invariant or varying?

invariant. as a shift in time doesn't affect the answer.

↳ eg. + shifted by a , results in the same signal

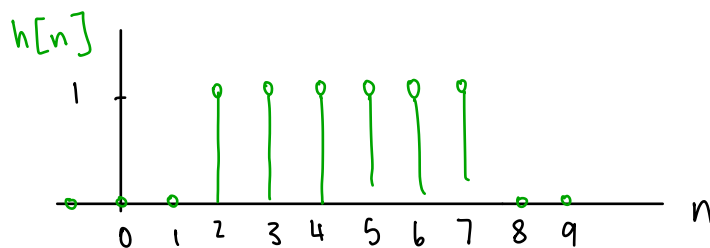
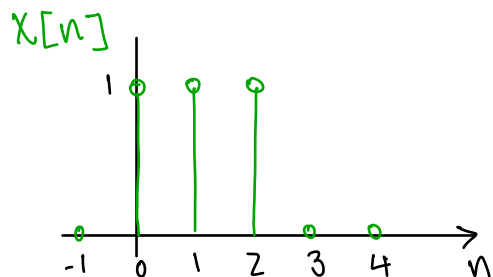
if + wasn't shifted

↳ eg. $y[n] = \frac{1}{2}x[n] + 3x[n-1]$

$$y[n-a] = \frac{1}{2}x[n-a] + 3x[n-1-a]$$

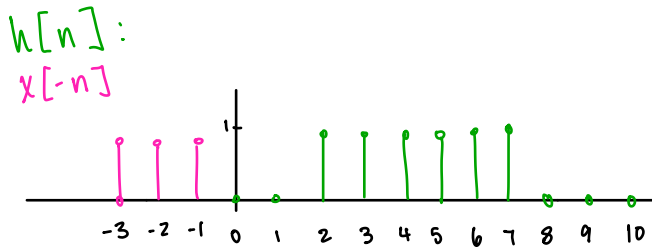
$$y[n] = y[n-a]$$

- (25) 3. Consider the discrete-time signals $x[n] = u[n] - u[n - 3]$ and $h[n] = u[n - 2] - u[n - 8]$, where $u[n]$ is the discrete time unit step function.
- (8) (a) Sketch a stem plot of $x[n]$ and $h[n]$. Make sure you label the values.



- (17) (b) Compute and sketch a stem plot of the convolution $y[n] = x[n] * h[n]$. Make sure you label the values.

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$



shifts:

• no initial overlap:

• $k \leq 1$: $y[k] = 0$

• $k=2$: $y[2] = 1$

$k=3$: $y[3] = 2$

$k=4$: $y[4] = 3$

$k=$: $y[5] = 3$

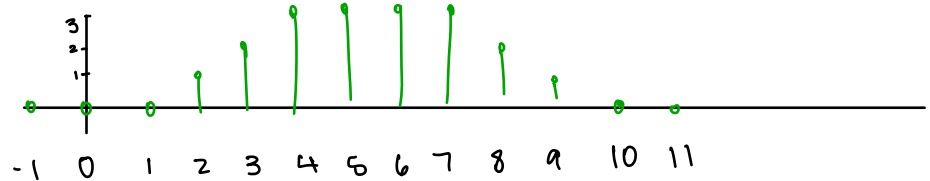
$y[6] = 3$

$y[7] = 2$

$y[8] = 1$

$k=10$: $y[10] = 0$

$y[11] = 0$



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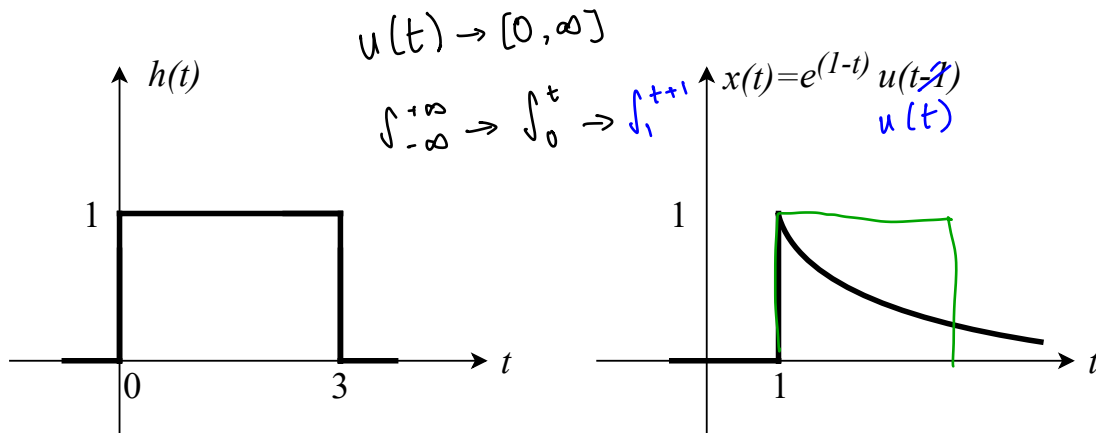
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- (8) (b) For what range(s) of values of t is $c(t)$ equal to zero?

When $t \leq 1$ Only.

- (8) (c) Write out the integral(s) (including the integration limits) that can be evaluated in order to find $c(t)$ for $1 \leq t < 4$. Do not evaluate the expression. Draw a box around your final answer.

$$\int_1^t e^{(1-\tau)} d\tau \text{ OR } \int_0^{t-1} e^{-(t-\tau-1)} d\tau$$

- (8) (d) Write out the integral(s) (including the integration limits) that can be evaluated in order to find $c(t)$ for $t \geq 4$. Do not evaluate the expression. Draw a box around your final answer.

$$\int_{t-3}^t e^{(1-\tau)} d\tau \text{ OR } \int_0^3 e^{-(t-\tau-1)} d\tau$$

- (10) 2. Determine whether the following discrete-time system is causal or noncausal, have memory or is memoryless, is linear or nonlinear, is time invariant or time varying. Justify your answers. In the following system, $x[n]$ is an arbitrary input and $y[n]$ is the response to $x[n]$.

$$y[n] = \frac{1}{2}x[n] + 3x[n-1]$$

(1 mark) The system is Causal: (1 mark) since the output is expressed only in the terms of current or past versions of the input.

(1 mark) The system has Memory: (1 mark) since the current output includes a past version of the input.

(1 mark) The system is Linear: (2 marks) for the justification.

$$y_1[n] = \frac{1}{2}x_1[n] + 3x_1[n-1]$$

$$y_2[n] = \frac{1}{2}x_2[n] + 3x_2[n-1]$$

$$\text{Let } x[n] = a_1x_1[n] + a_2x_2[n]$$

$$\text{then } y[n] = \frac{1}{2}(a_1x_1[n] + a_2x_2[n]) + 3(a_1x_1[n-1] + a_2x_2[n-1])$$

$$\text{rearrange } y[n] = a_1(\frac{1}{2}x_1[n] + 3x_1[n-1]) + a_2(\frac{1}{2}x_2[n] + 3x_2[n-1])$$

$$\text{then } y[n] = a_1y_1[n] + a_2y_2[n]$$

(1 mark) The system is Time-Invariant: (2 marks) for the justification.

Let the input be $x_1[n]$.

$$\text{Therefore } y_1[n] = \frac{1}{2}x_1[n] + 3x_1[n-1].$$

Now let the input be $x_2[n] = x_1[n-N]$, where N is an integer.

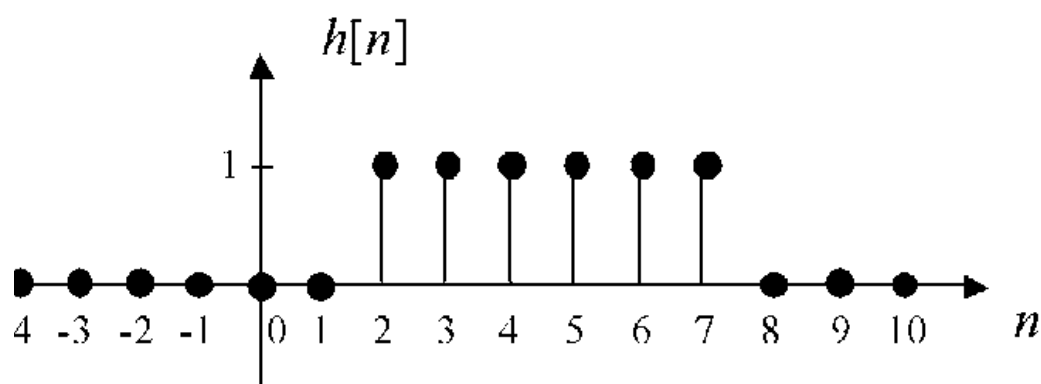
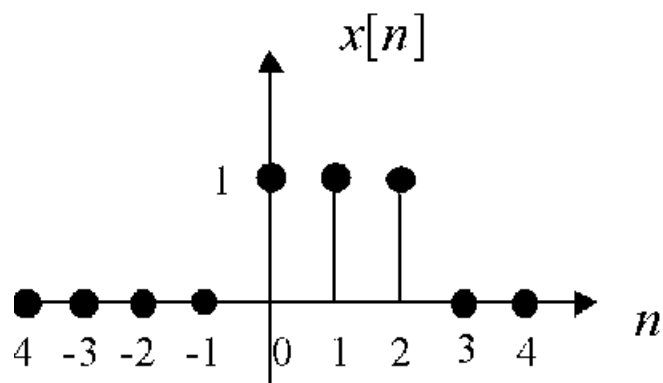
$$\text{The output will now be } y_2[n] = \frac{1}{2}x_2[n] + 3x_2[n-1]$$

$$\text{Or } y_2[n] = \frac{1}{2}x_1[n-N] + 3x_1[n-1-N]$$

$$\text{Hence, } y_2[n] = y_1[n-N].$$

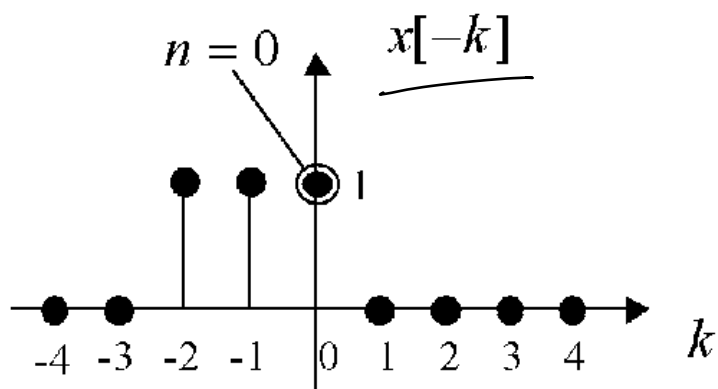
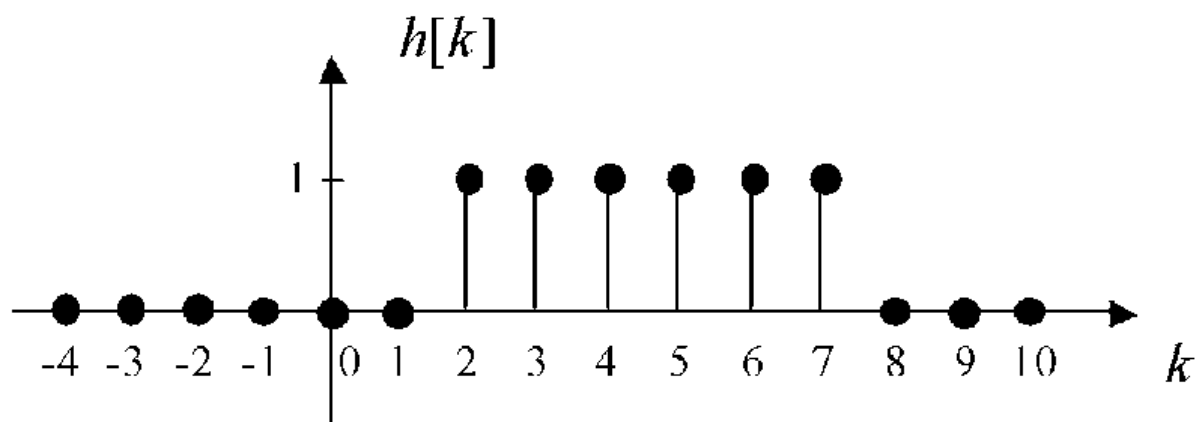
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- (17) (b) Compute and sketch a stem plot of the convolution $y[n] = x[n] * h[n]$. Make sure you label the values.

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$



$$\begin{aligned}
& n < 2, \quad h[k]x[n-k] = 0 \quad \forall k, \quad y[n] = 0 \\
& y[2] = (1)(1) = \textcircled{1} \\
& y[3] = (1)(1) + (1)(1) = 2 \\
& y[4] = (1)(1) + (1)(1) + (1)(1) = 3 \\
& y[5] = (1)(1) + (1)(1) + (1)(1) = 3 \\
& y[6] = (1)(1) + (1)(1) + (1)(1) = 3 \\
& y[7] = (1)(1) + (1)(1) + (1)(1) = 3 \\
& y[8] = (1)(1) + (1)(1) = 2 \\
& y[9] = (1)(1) = 1 \\
& n > 10, \quad h[k]x[n-k] = 0 \quad \forall k, \quad y[n] = 0
\end{aligned}$$

